

Formal Methods for Software Engineering homework report

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Table of Contents

1. Design choices	2
1.1. World movement	2

1. Design choices

1.1. World movement

Since all elements of a row move with the same speed, to reduce the state-space size, we decided to make the whole row move with that speed: we declared a matrix containing the initial state of the world, and an array of offsets, one per row. Then, every time the world's clock ticks, for each row we check if the number of ticks is a multiple of the row's speed and, if so, we increase the offset of that row. To get the current element in a cell, we just need to look at the matrix in the cell at the given x **minus** the offset multiplied by the direction of the row (one for left-to-right movement, -1 for right-to-left movement): let's assume the row moves from left to right, and we have an offset of 1. What it means is that the current cell corresponding to x is actually the cell just before, as the whole row just shifted one to the right.